



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SKS AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

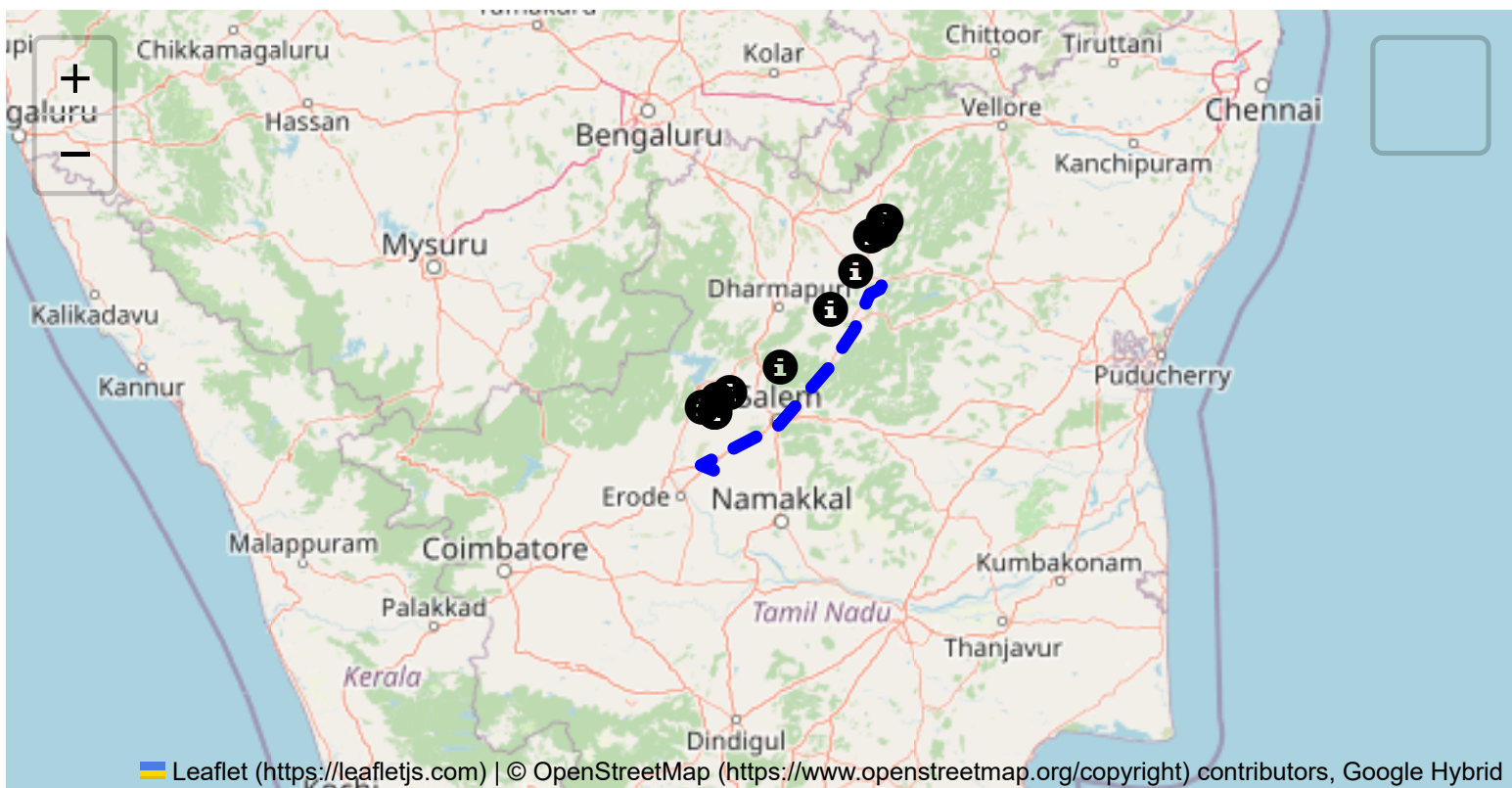
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

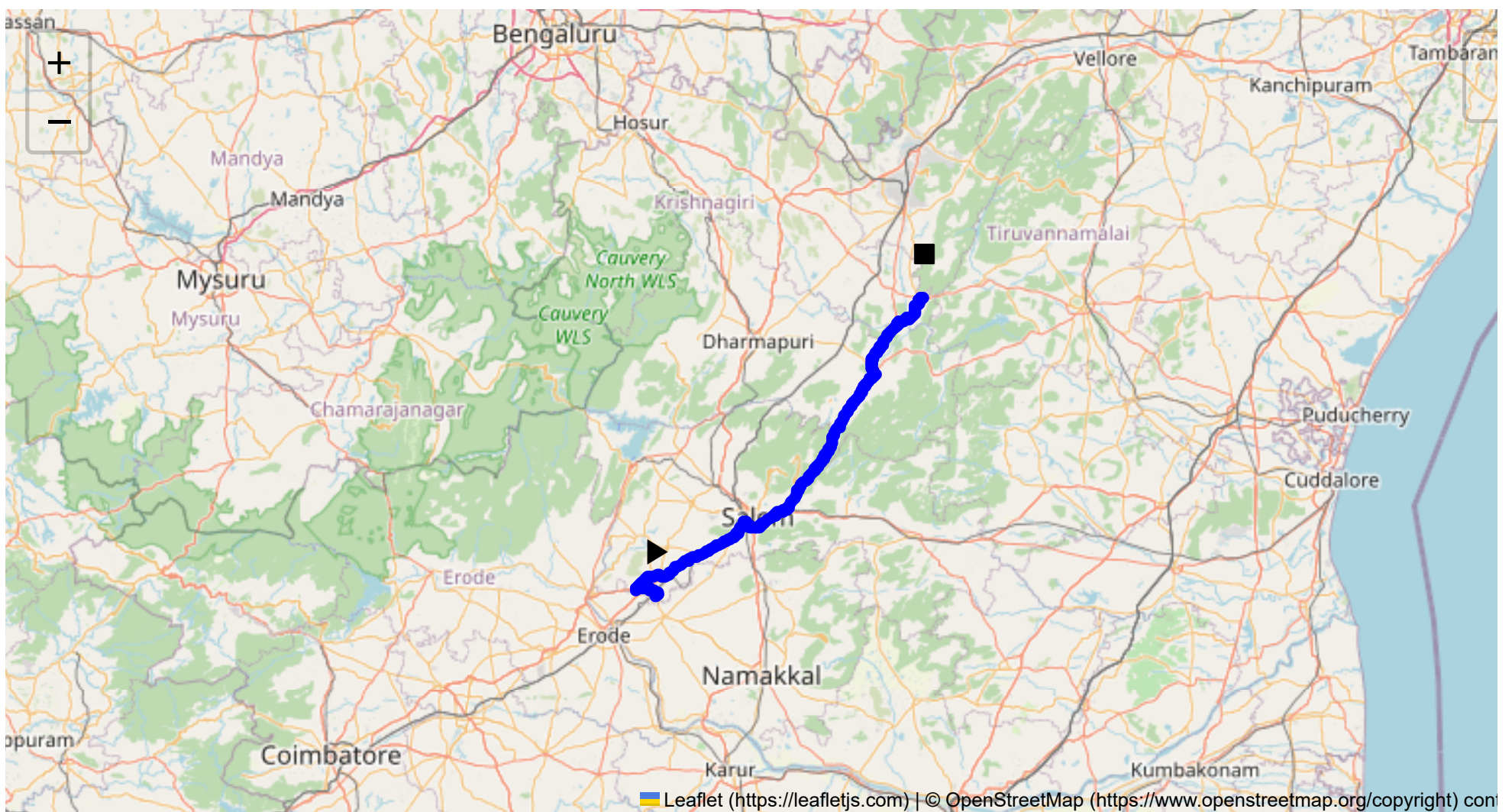
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 147.45 km
Estimated Duration: 3.5 hours
Adjusted Duration (Heavy Vehicle): 4.4 hours
Start: (11.4381, 77.8734)
End: (12.25333319, 78.6199476)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Kurungapatti covers approximately 147.45 kilometers and passes through various regions in Tamil Nadu, utilizing major roads such as the Salem - Kochi Highway, Salem By Pass, and NH 179A. This journey involves traversing urban, semi-urban, and rural landscapes.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences tropical weather, with hot and humid summers, a intense monsoon season from June to September, and a cool, dry season in the winter. The monsoon season can lead to heavy rainfall, causing potential flooding, slippery roads, and reduced visibility. It's crucial to monitor weather forecasts during these months.

3. Analysis of Traffic Patterns

Major congestion points can be expected near K R Nagar and along the Salem By Pass, especially during morning and evening rush hours (8:00 AM - 10:00 AM and 5:00 PM - 8:00 PM). This is due to both local traffic and transit vehicles. Moderate to heavy traffic can occur at agriculture and commercial hubs, particularly in Salem.

4. Assessment of Road Quality and Infrastructure

Overall, the road quality varies. The highways are generally well-maintained and suitable for heavy vehicles. However, smaller roads, especially through rural areas or villages, may have potholes, uneven surfaces, or lack proper signage, posing potential risks for heavy and/or oversized vehicles.

5. Suggestions for Alternative Routes for Emergencies

In case of emergencies, alternative routes can include diversions onto parallel highways or state roads intersecting the primary route. However, these might not be suitable for all vehicle types due to weight restrictions or narrower lanes. It's advisable to have real-time navigation aids ready.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu follows strict guidelines for the transport of hazardous materials, focusing on proper labeling, documentation, and transport permits. It's necessary to adhere to designated routes and operate within specific hours to minimize risk, mainly during low traffic periods.

7. Overview of Historical Incidents

Historically, collisions or rollovers involving heavy vehicles have occurred mostly near congested urban areas or sharp curves on highways. These incidents often arise from over-speeding or poor road conditions during rainy weather.

8. Environmental Considerations and Sensitive Areas

Sensitive areas include agricultural lands and some protected forest patches near Salem. Avoiding contamination from spills is critical, as these areas are environmentally significant.

9. Analysis of Communication Coverage

Communication networks are generally strong along the route, especially near urban centers. However, dead zones might occur in more remote or less developed regions, particularly in rural stretches.

10. Estimated Emergency Response Times

Emergency response times in urban areas like Salem are quicker (within 30 minutes) due to proximity to services. Rural areas might experience delays ranging from 45 minutes to over an hour due to distance and road access.

12. Overall Summary of Risk Assessment

The journey from Sangagiri to Kurungapatti presents moderate levels of risk, primarily affected by weather conditions, traffic congestion in urban areas, and variable road quality. Strict adherence to regulations and awareness of alternative routes can mitigate these risks. Preparing for monsoon-related disruptions and keeping open communication channels can help in effectively managing potential hazards during transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43961, 77.87341	15 KM/Hr
2	Turn	High	11.44024, 77.87528	15 KM/Hr
3	Turn	High	11.44898, 77.87410	15 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45764, 77.85219	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45545, 77.81812	30 KM/Hr
8	Turn	High	11.45584, 77.81695	15 KM/Hr
9	Turn	Medium	11.45656, 77.81709	30 KM/Hr
10	Turn	Medium	11.53552, 77.95763	30 KM/Hr
11	Turn	Medium	11.64096, 78.11795	30 KM/Hr
12	Turn	High	11.64210, 78.11860	15 KM/Hr
13	Turn	High	11.64172, 78.12154	15 KM/Hr
14	Turn	Medium	11.64098, 78.12163	30 KM/Hr
15	Turn	High	11.63972, 78.12083	15 KM/Hr
16	Turn	Medium	11.63668, 78.12232	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
17	Turn	Medium	11.62826, 78.13337	30 KM/Hr
18	Turn	High	11.66372, 78.20649	15 KM/Hr
19	Turn	Medium	11.67124, 78.22961	30 KM/Hr
20	Turn	High	11.67111, 78.23134	15 KM/Hr
21	Turn	High	11.67187, 78.23142	15 KM/Hr
22	Turn	Medium	11.67427, 78.23892	30 KM/Hr
23	Turn	Medium	11.81208, 78.34071	30 KM/Hr
24	Turn	Medium	11.83518, 78.35305	30 KM/Hr
25	Turn	Medium	11.84832, 78.36115	30 KM/Hr
26	Turn	Medium	12.04286, 78.48352	30 KM/Hr
27	Turn	Medium	12.05026, 78.48298	30 KM/Hr
28	Turn	Medium	12.05407, 78.48313	30 KM/Hr
29	Turn	Medium	12.07570, 78.47610	30 KM/Hr
30	Turn	Medium	12.17756, 78.54790	30 KM/Hr
31	Turn	Medium	12.17838, 78.54853	30 KM/Hr
32	Turn	Medium	12.18684, 78.54965	30 KM/Hr
33	Turn	Medium	12.18752, 78.55009	30 KM/Hr
34	Turn	Medium	12.18883, 78.55864	30 KM/Hr
35	Turn	Medium	12.19710, 78.57531	30 KM/Hr
36	Turn	High	12.19786, 78.57624	15 KM/Hr
37	Turn	Medium	12.20375, 78.59119	30 KM/Hr
38	Turn	Medium	12.22486, 78.59702	30 KM/Hr
39	Turn	Medium	12.22650, 78.59653	30 KM/Hr
40	Turn	High	12.23464, 78.60023	15 KM/Hr
41	Turn	Medium	12.23635, 78.60528	30 KM/Hr
42	Turn	High	12.25218, 78.61196	15 KM/Hr
43	Turn	Medium	12.25381, 78.61757	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
12	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
14	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
15	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
16	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
17	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
18	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
19	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
20	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
21	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
22	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
24	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
25	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
26	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium
33	hospital	Nanjaya Hospital	12.0550888, 78.4821929	30 km/h	Medium
34	hospital	Government Hospital, Harur	12.061178, 78.4756913	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
35	hospital	Subam Hospital	12.0595128, 78.4766713	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
13	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
23	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
27	school	Indian Matriculation Higher Secondary school	12.0061755, 78.4533388	30 km/h	Medium
28	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
29	school	Chinnan Kuppam Government High School	12.0190729, 78.4581955	30 km/h	Medium
30	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
31	school	Jayam Vidhyalaya Matriculation Higher Secondary School	12.0264103, 78.4656553	30 km/h	Medium
32	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45545, 77.81812



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45584, 77.81695



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45656, 77.81709



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53552, 77.95763



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64096, 78.11795



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64210, 78.11860



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64172, 78.12154



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64098, 78.12163



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63972, 78.12083



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63668, 78.12232



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62826, 78.13337



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66372, 78.20649



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67124, 78.22961



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67111, 78.23134



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67187, 78.23142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67427, 78.23892



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.81208, 78.34071



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83518, 78.35305



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84832, 78.36115



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04286, 78.48352



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.05026, 78.48298



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.05407, 78.48313



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.07570, 78.47610



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17756, 78.54790



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17838, 78.54853



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.18684, 78.54965



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.18752, 78.55009



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.18883, 78.55864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19710, 78.57531



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.19786, 78.57624



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.20375, 78.59119



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.22486, 78.59702



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.22650, 78.59653



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.23464, 78.60023



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.23635, 78.60528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.25218, 78.61196



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25381, 78.61757
